to use Bloom filters and giving them the instruments to do so.

Starting with section 4.8, the focus shifts to theory, providing a background for the interested reader that wants to understand not just how, but why Bloom filters work. To facilitate understanding of these sections, readers can go through appendix F (as well as appendices B and C, if needed) to review randomized algorithms, big-o notation and basic data structures, if needed, prior to starting with section 4.8.

The next few sections will then closely examine the performance of our data structure, both intended as running time and memory print (section 4.9) and accuracy of the algorithm (4.10).

Finally, section 4.11 describes some of the most advanced variants of Bloom filters used to provide new features or lower the false-positive rate.

4.1 The Dictionary Problem: Keeping Track of Things

Let’s go over a hypothetical scenario. You work for a company that is large enough to maintain its own email service. This is a legacy service, providing basic features only. After the last reorganization, the new CTO decides that you need to reinvent it to be aggressive on the market segment, and she puts your new manager in charge of the product redesign.